One more exposure

01

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Separate a sighting from a prediction.

CHOOSE ONE JOB TOGETHER

TELESCOPE SCHEDULER ☐

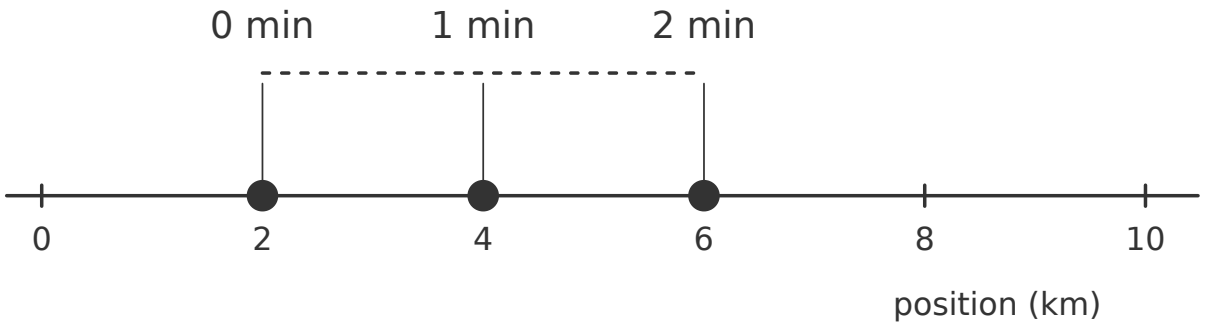
Mark where to aim at minute **3**. Tell the operator which dots were measured and which dot you predicted.

SCIENCE BROADCASTER ☐

Give viewers a two-sentence update. Point to a measured dot, then say where you expect the next one.

Word help: sighting / prediction / pattern

Three telescope fixes are at **2**, **4** and **6 km** at minutes **0**, **1** and **2**. The night crew has one more exposure.



01 OUR FIRST CALL

Where should the camera look next? Use the timed dots in your call.

Aim at ___ because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new sighting fit your prediction? Tell the next crew what to check.

Now we would say ___ because ___.

Switch speakers. What new observation would make you change your prediction? Explain why.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Send the rover home

02

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Tell distance traveled and change in position apart.

CHOOSE ONE JOB TOGETHER

ROUTE ENGINEER ☐

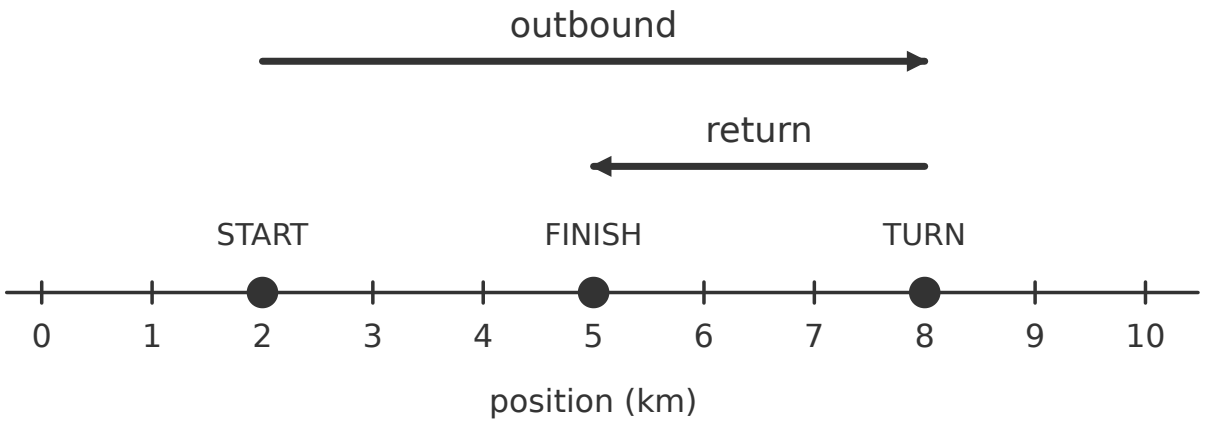
Mark both parts of the route and draw one start-to-finish arrow. Report the trip length and change in position.

RADIO DISPATCHER ☐

Guide a new driver to the finish. Give the trip length, change in position, and final location.

Word help: route / distance / displacement

A survey rover tracks XJ's model from **+2** to **+8** to **+5 km**. Dispatch needs both mileage and final location.



01 OUR FIRST CALL

What numbers should dispatch use? Say what each number measures.

The trip was ___. The finish is ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the new location instruction. What should dispatch tell the driver now?

Now we would say ___ because ___.

Switch speakers. If right becomes negative, which numbers change sign? Which quantities stay the same?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Is it really slowing down?

03

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a position-time graph to find velocity.

CHOOSE ONE JOB TOGETHER

RADAR OPERATOR ☐

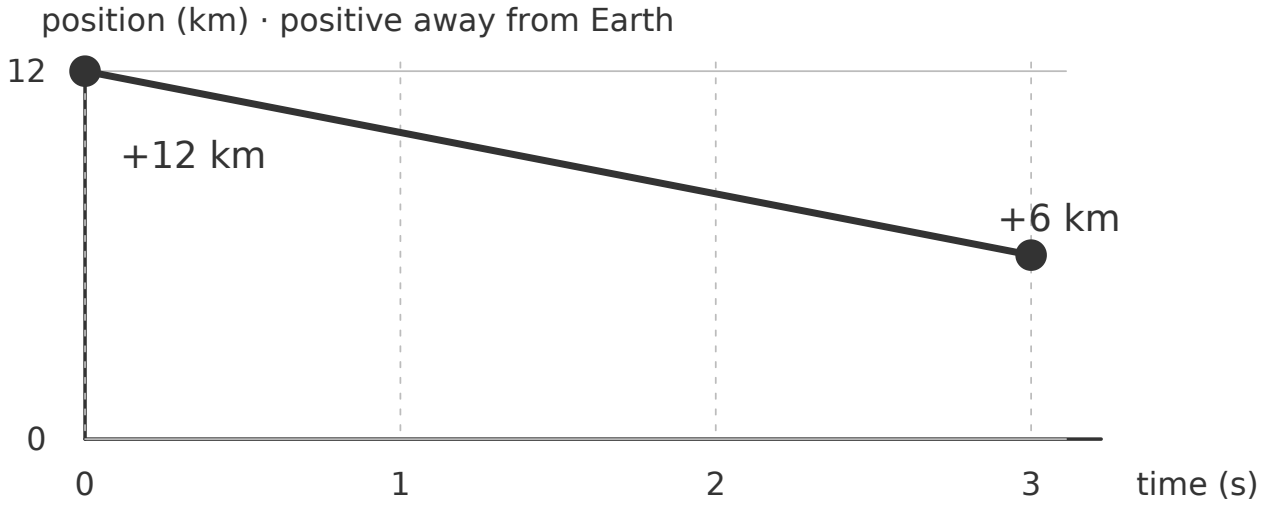
Calculate the signed average velocity. Mark the two sightings on the graph so the next operator can check your result.

TRAFFIC BROADCASTER ☐

Report a speed and direction. Tell viewers whether these two sightings show that XJ is slowing down.

Word help: position / velocity / toward

Radar sees XJ at **+12 km**, then **+6 km** three seconds later. Positive points away from Earth.



01 OUR FIRST CALL

What can you report about XJ's motion? Use the graph to support your call.

XJ moves ___ at an average of ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the new caption. Write the correction you would send or say on air.

Now we would say ___ because ___.

Switch speakers. What can these two sightings tell you about average velocity? What can they not tell you about motion between sightings?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Where does the camera finish?

04

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use signed graph areas to find displacement.

CHOOSE ONE JOB TOGETHER

TRACKING ENGINEER ☐

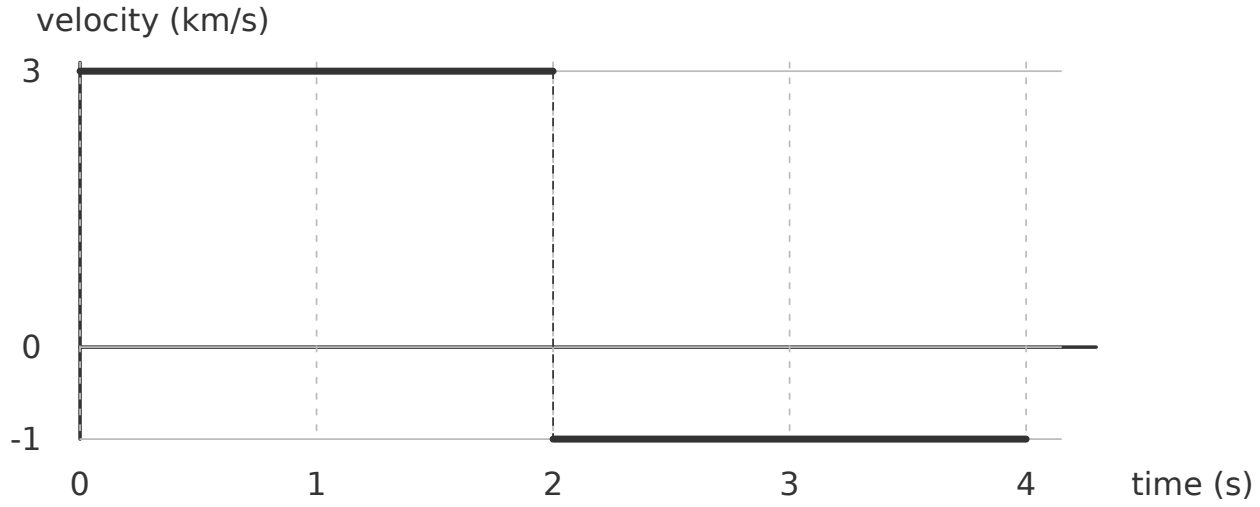
Shade both parts of the velocity graph. Radio the camera's final change in position, including direction.

REPLAY PRODUCER ☐

Mark the outward and return shots. Report total distance and net movement to the editor.

Word help: area / return / displacement

A moving camera follows a model at **+3 km/s** for **2 s**, then **–1 km/s** for **2 s**. The editor needs the end location and the total footage distance.



01 OUR FIRST CALL

Where does the camera finish relative to its start? How far did it travel?

The camera travels ___ but ends ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the new caption. Which number belongs on the location graphic? Explain.

Now we would say ___ because ___.

Switch speakers. If the camera also moves sideways, why does that motion need its own velocity-time graph?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

How wide should we search?

05

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Read forecast bounds without inventing a probability.

CHOOSE ONE JOB TOGETHER

SURVEY ENGINEER ☐

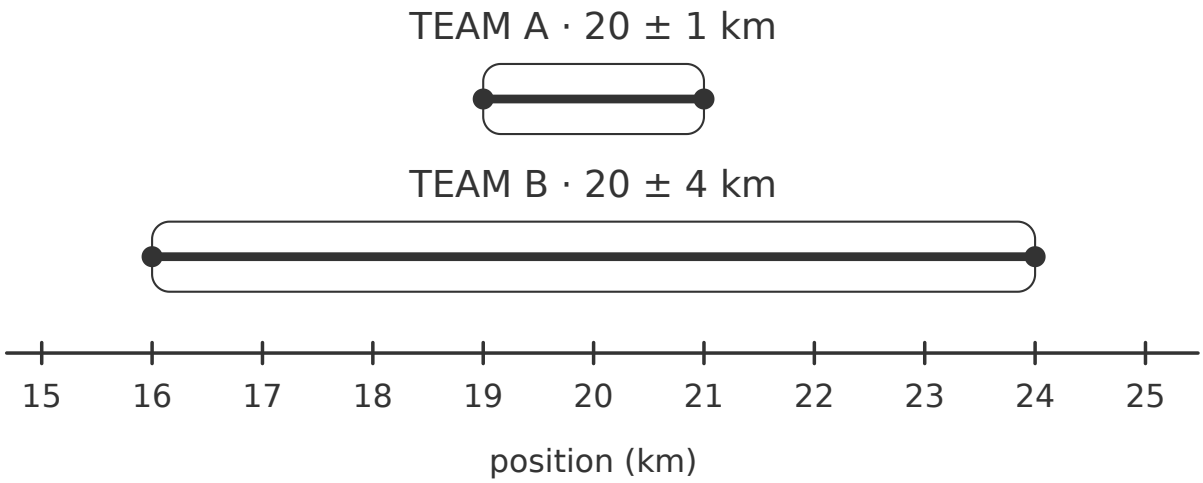
Label both search bands with their endpoints. Recommend which report needs a wider telescope search.

WEATHER-DESK PRESENTER ☐

Present both forecast bands. Explain which search is wider and what the bands do not tell us about impact.

Word help: center / range / endpoints

Two XJ teams predict a center of **20 km**. Team A gives **±1 km**; team B gives **±4 km**. A telescope has one narrow field.



01 OUR FIRST CALL

What search area should the crew prepare for? Use both sets of endpoints.

Team ___ needs a wider search because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the editor's new label. What can you report from these bands?

Now we would say ___ because ___.

Switch speakers. How is changing an input's uncertainty different from changing how you expect XJ to move?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Send the whole movement

06

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Combine two displacement components into one arrow.

CHOOSE ONE JOB TOGETHER

NAVIGATION ENGINEER ☐

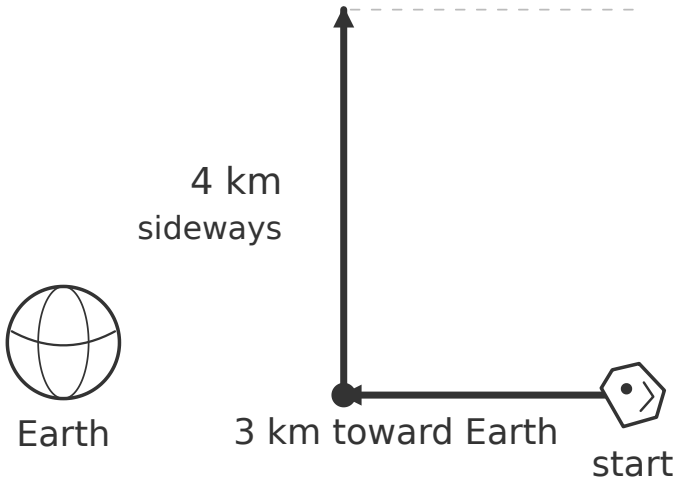
Draw one arrow for the whole displacement. Give its length and direction, and label both component legs.

MAP PRESENTER ☐

Use the map to report the whole movement. Explain why '**7 km** closer to Earth' is the wrong caption.

Word help: component / sideways / resultant

A tracker sees a **3 km** displacement toward Earth and **4 km** sideways. The second telescope needs the whole movement.



01 OUR FIRST CALL

What full movement should the second telescope track? Use your map.

The whole displacement is ___ toward ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What part can the updated tracker measure? What part will it miss?

Now we would say ___ because ___.

Switch speakers. Give the direction as an angle. Name the line you measure that angle from.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Choose the speed alert

07

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a velocity change to find acceleration.

CHOOSE ONE JOB TOGETHER

TELEMETRY ENGINEER ☐

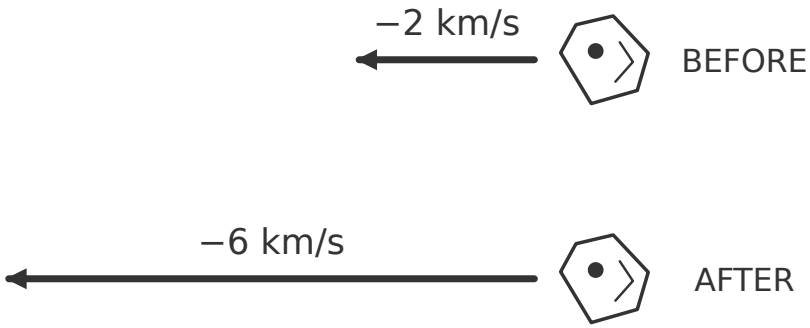
Radio the average acceleration with its sign and unit. Label both velocity arrows so the crew can check it.

REPLAY COMMENTATOR ☐

Narrate the two-second change. Point to the arrow lengths to explain whether XJ sped up or slowed down.

Word help: velocity / acceleration / speeding up

Two signed XJ velocities are **−2** and **−6 km/s**, two seconds apart. The control room must choose its speed alert.



01 OUR FIRST CALL

Which speed alert should the control room use? Support it with the arrows.

The speed ___ because the arrow ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the proposed banner. Give a corrected report using the two velocities.

Now we would say ___ because ___.

Switch speakers. Why do two velocity readings give an average acceleration rather than the acceleration at every instant?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Catch the next shot

08

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a velocity-time graph to predict displacement.

CHOOSE ONE JOB TOGETHER

CAMERA ENGINEER ☐

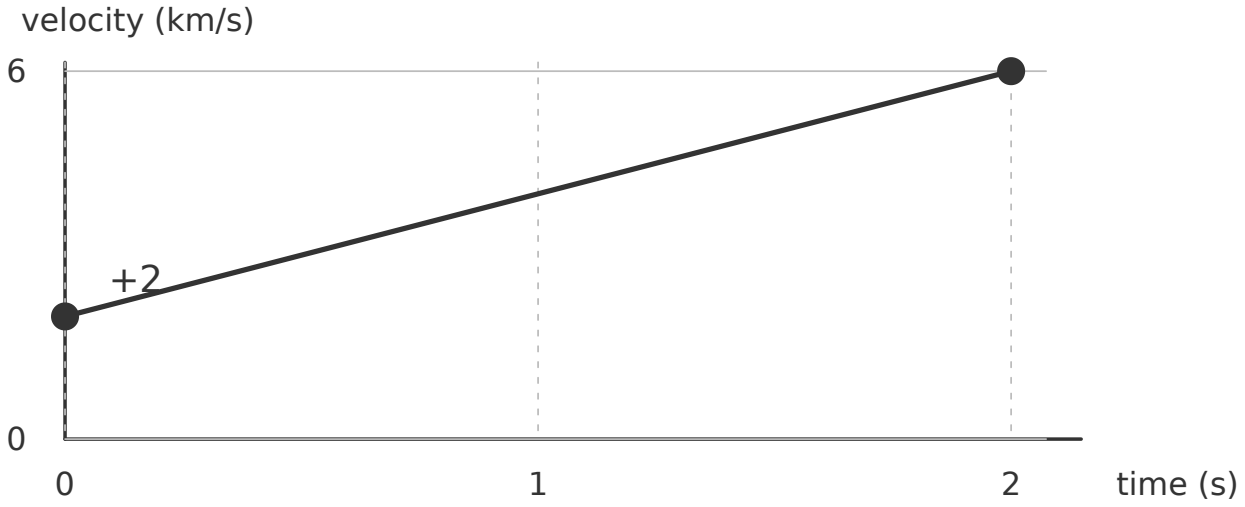
Shade the area under the velocity-time trace. Send the camera's displacement at the end of the shot.

REPLAY EDITOR ☐

Explain why using the final speed for the whole clip would miss the shot. Point to the graph as you speak.

Word help: area / constant / displacement

XJ moves from **+2** to **+6 km/s** over **2 s**. The rehearsal assumes constant acceleration. Where should the next camera look relative to the start?



01 OUR FIRST CALL

Where should the camera look relative to the start? Show how the graph helps.

Our forecast is ___ if ___ stays constant.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new report still let you guarantee that displacement? Explain your revised call.

Now we would say ___ because ___.

Switch speakers. Check your first answer with a kinematics equation. What does that equation treat as constant?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

One camera. Which window?

09

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use timed sightings to make a prediction, then check it.

CHOOSE ONE JOB TOGETHER

TRACKING ENGINEER ☐

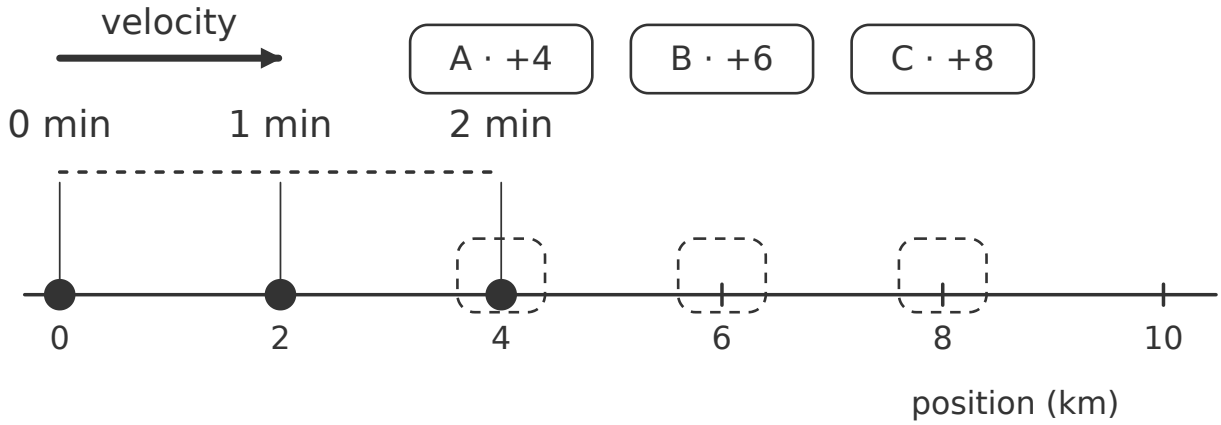
Radio a camera window. Point to the sightings that support your call.

SCIENCE BROADCASTER ☐

Give a **15**-second update. Use your cart drawing to explain why XJ can keep moving without an engine.

Word help: aim / window / coast

XJ is drifting right. At minute **3**, one camera can aim at one window: A, B or C.



01 OUR FIRST CALL

Where should the camera look? Include a signed velocity and say what you expect to stay the same.

Aim at ___ because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

How far did XJ move in the last minute? Use that result in your revised call.

Now we would say ___ because ___.

Switch speakers. Why could a straight-line forecast work for a minute even while gravity pulls on XJ?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Who is doing the pushing?

10

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Name the source and receiver of each force.

CHOOSE ONE JOB TOGETHER

SIMULATION ENGINEER ☐

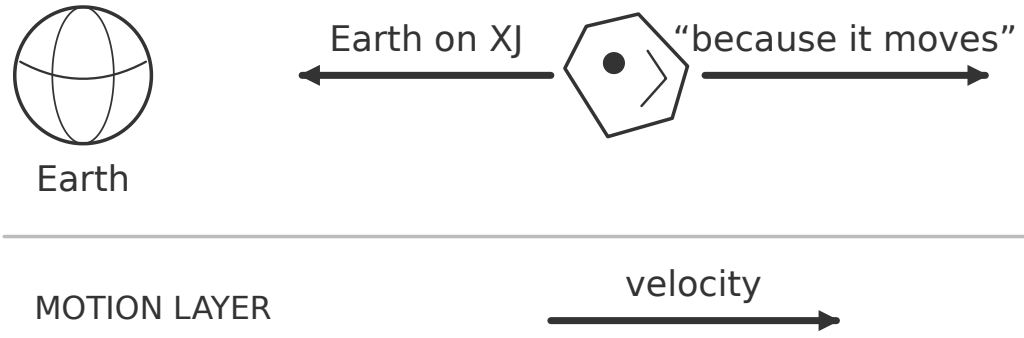
Approve the force layer for XJ. Name who exerts each force and what receives it. Keep motion separate.

SCIENCE GRAPHICS EDITOR ☐

Prepare the viewer graphic. Ask your partner who causes each proposed force before you approve its label.

Word help: source / receiver / gravity

An XJ animation shows gravity from Earth and a forward arrow labeled “because it moves.” The graphics desk needs approved force labels.



01 OUR FIRST CALL

Which force arrows should go live? Explain each arrow you keep or remove.

Keep ___ because ___ acts on ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new information change the force layer? Explain your update.

Now we would say ___ because ___.

Switch speakers. Name another real source of force on XJ. Where would its arrow go, and what would you label it?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Same push. Different mass.

11

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare velocity changes when masses differ.

CHOOSE ONE JOB TOGETHER

TEST ENGINEER ☐

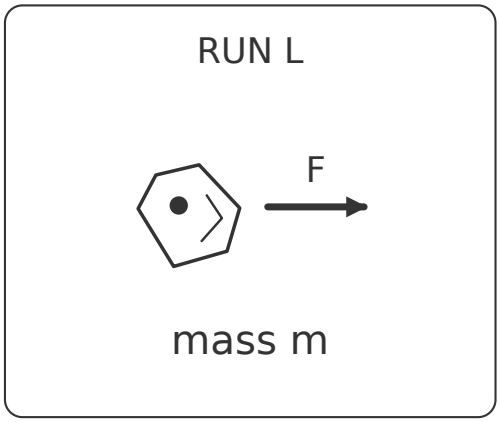
Compare the two test runs. Tell the crew which body changes velocity more and use the masses to explain why.

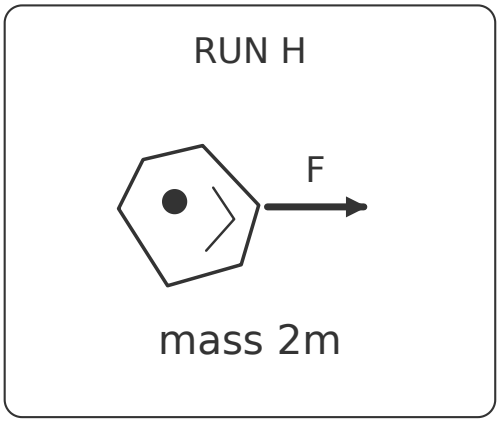
LAB CORRESPONDENT ☐

Explain why equal pushes can give different results. Have your partner check what stayed the same in both tests.

Word help: mass / net force / velocity change

Two asteroid models get the same net force for the same time. Model H has twice the mass of L.





01 OUR FIRST CALL

Which body changes velocity more? Explain your first call from the test card.

Run ___ changes more because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What changed in the test? Give the crew a new comparison.

Now we would say ___ because ___.

Switch speakers. Check both runs using mass times acceleration. Do your results match the forces on the test card?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Which run changes more?

12

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare velocity changes when net forces differ.

CHOOSE ONE JOB TOGETHER

FLIGHT-TEST ENGINEER ☐

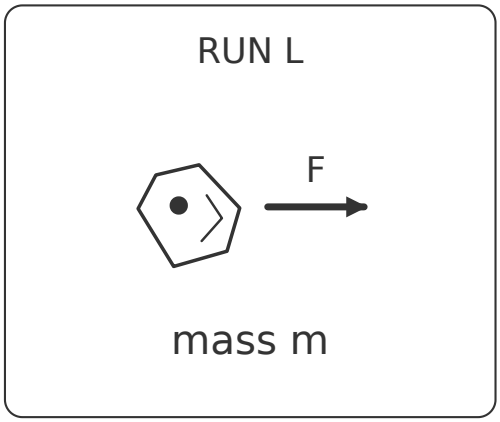
Add velocity-change arrows for both runs. Recommend the larger change and name what stayed the same.

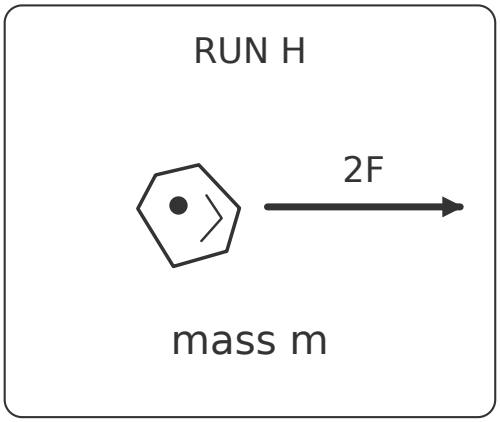
TEST-RANGE BROADCASTER ☐

Tell viewers which run changes velocity more. Ask whether doubling motor pull always doubles the net force.

Word help: net force / drag / velocity change

A rehearsal uses the same asteroid model twice. The second net push is twice the first, for the same time. Operations needs a comparison.





01 OUR FIRST CALL

How do the two velocity changes compare? Point to the force arrows.

The velocity change is ___ if ___ stays the same.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Do the new readings still support your comparison? Explain what you must check.

Now we would say ___ because ___.

Switch speakers. Explain the difference between doubling one applied force and doubling the net force.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Moving left. Pushed right.

13

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Distinguish motion, net force, and acceleration.

CHOOSE ONE JOB TOGETHER

CONTROLS ENGINEER ☐

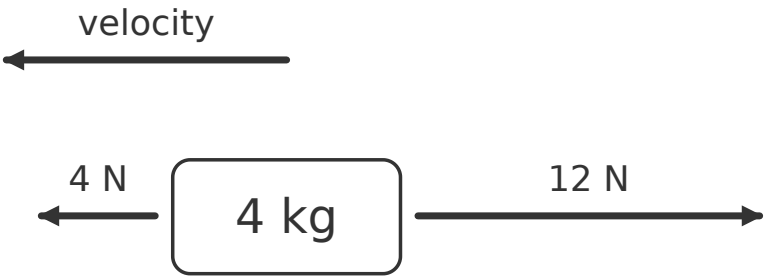
Draw the net-force and acceleration arrows. Tell the operator what happens to speed while the body still moves left.

REPLAY ANALYST ☐

Label separate arrows for motion and acceleration. Explain to viewers how they can point opposite ways.

Word help: net force / acceleration / slowing down

A tracking model moves left. Its **4 kg** body feels **12 N** right and **4 N** left. The status display has separate motion and force arrows.



01 OUR FIRST CALL

What happens to speed at first? Include the net force and acceleration in your call.

It moves ___ while acceleration points ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the updated display. What does a force arrow tell you, and what does it not tell you?

Now we would say ___ because ___.

Switch speakers. What additional information would let you predict when the body stops and reverses?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Two pulls. Two receivers.

14

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Place an interaction pair on its two receivers.

CHOOSE ONE JOB TOGETHER

SIMULATION ENGINEER ☐

Put each gravitational arrow on the body that receives it. Send the animator two corrected object panels.

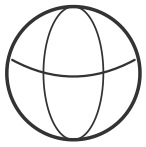
SCIENCE GRAPHICS PRODUCER ☐

Write a split-screen caption naming who pulls whom. Explain why equal force does not require equal acceleration.

Word help: interaction / receiver / equal and opposite

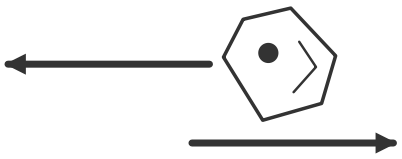
Earth pulls on XJ. A public animation shows both gravitational arrows on XJ, with the caption “they cancel.”

EARTH PANEL



Earth

XJ PANEL



PROPOSAL: “They cancel.”

01 OUR FIRST CALL

Do the two arrows cancel on XJ? Explain using your corrected panels.

___ pulls on ___. That arrow belongs on ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Which arrow belongs in the updated shot? Name its source and receiver.

Now we would say ___ because ___.

Switch speakers. If Earth and XJ are one combined system, which forces in this interaction are internal to that system?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Build the force overlay

15

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Keep only forces on the chosen object in its diagram.

CHOOSE ONE JOB TOGETHER

SYSTEMS ENGINEER ☐

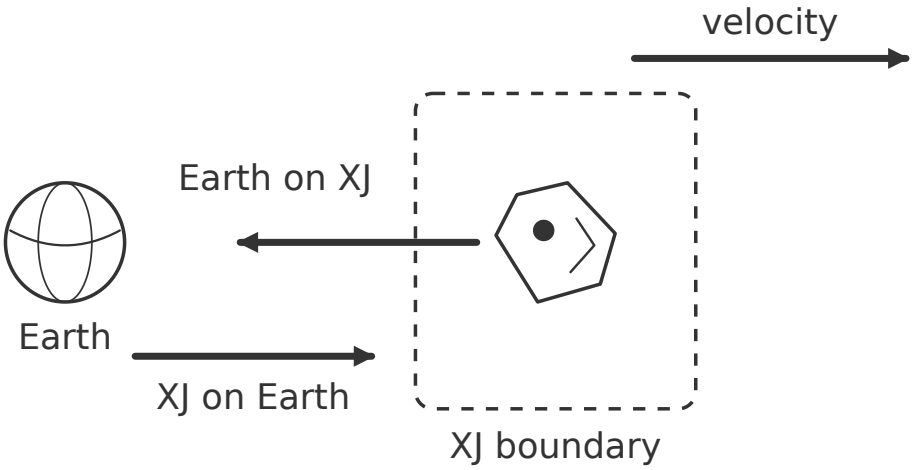
Outline XJ as the chosen object. Keep only the force arrows you would use to calculate XJ's acceleration.

ANIMATION DIRECTOR ☐

Separate the force and motion layers. Tell the animator which Earth arrow belongs in a different shot.

Word help: boundary / force / motion

The mission simulator needs one overlay called “forces on XJ.” Its proposed layers include Earth on XJ, XJ on Earth and XJ’s velocity.



01 OUR FIRST CALL

Which arrows belong in 'forces on XJ'? Explain what you keep and move.

This arrow acts on ___, so it belongs ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Where does the new arrow belong? Update the force layer and explain.

Now we would say ___ because ___.

Switch speakers. Must XJ move in the direction of its net force? Give a possible motion that shows your reasoning.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

One pull, two components

16

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Represent one force without counting it twice.

CHOOSE ONE JOB TOGETHER

NAVIGATION ENGINEER ☐

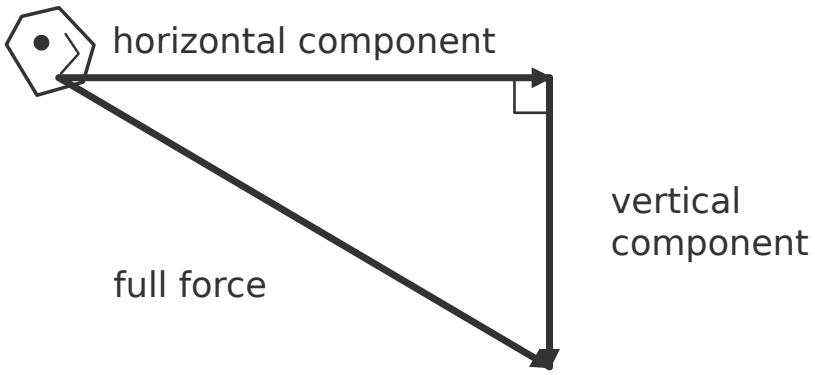
Prepare the component view for the calculator. Mark the full-force arrow that these components replace.

MAP GRAPHICS EDITOR ☐

Show the full force and its component view. Write a caption that prevents counting the same pull twice.

Word help: force / component / sum

A navigation graphic lists a gravitational force AND its two components as three separate pulls on XJ.



01 OUR FIRST CALL

Which arrows should enter the force sum? Explain your choice.

Use ___ because the components represent ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the calculator request. What correction would you send before it runs?

Now we would say ___ because ___.

Switch speakers. If XJ's velocity points horizontally, which force component changes its speed at that instant? Why?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Cut from the table to space

17

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use contact surfaces to decide when friction acts.

CHOOSE ONE JOB TOGETHER

EFFECTS ENGINEER ☐

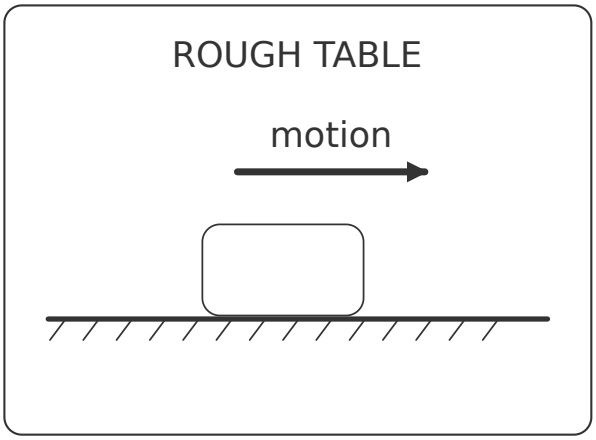
Set friction for each scene. Mark the touching surfaces and explain what could change XJ's motion in empty space.

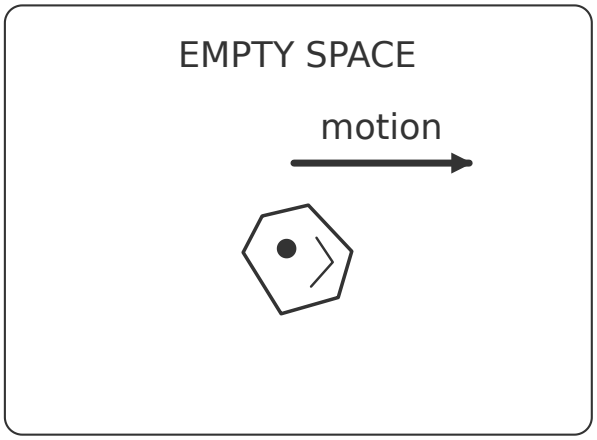
SCIENCE BROADCASTER ☐

Correct the caption for the two shots. Point to the surfaces and say what could still bend XJ's path.

Word help: contact / friction / drag

A show cuts from a block sliding on a rough table to XJ coasting in empty space. The proposed caption says “Both will stop from surface friction.”





01 OUR FIRST CALL

What should the two-scene caption say? Explain using the contact surfaces.

In this scene, ___ touches ___, so ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What changed in the new shot? Update the force explanation for XJ.

Now we would say ___ because ___.

Switch speakers. How would entering an atmosphere change the forces you include? Explain where the added force comes from.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Stopped or coasting?

18

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Distinguish zero acceleration from zero velocity.

CHOOSE ONE JOB TOGETHER

FLIGHT CONTROLLER ☐

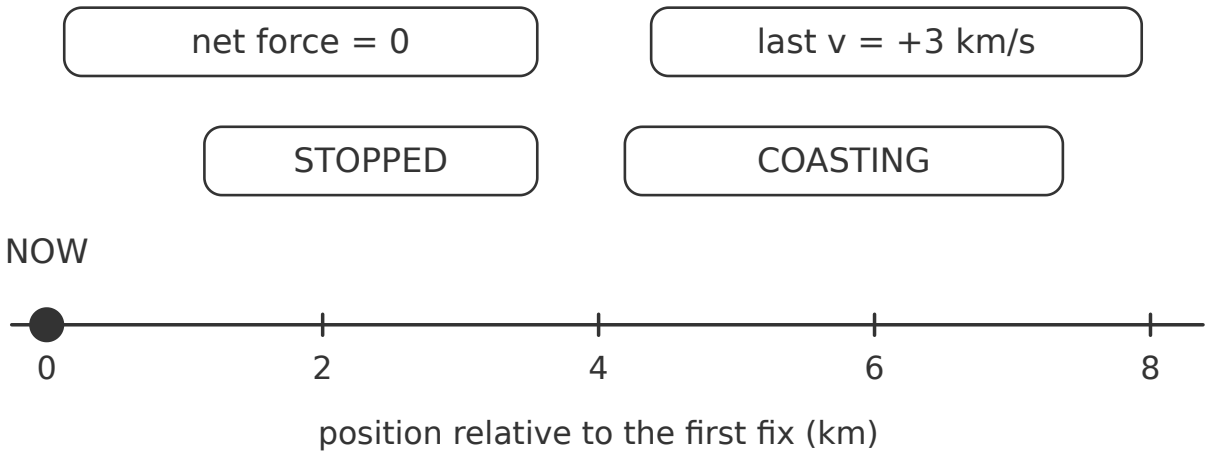
Choose the status graphic. Mark where XJ should be two seconds later relative to the first sighting.

BROADCAST PRODUCER ☐

Choose the on-air label. Explain why zero acceleration and zero speed do not mean the same thing.

Word help: balanced / coasting / acceleration

The flight display reads "net force zero." XJ's last velocity was **+3 km/s**. Two graphics say STOPPED and COASTING.



01 OUR FIRST CALL

Which status should go live? Use a second position to support your call.

The status is ___ because net force ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Can the old status describe the next interval? Explain what the new report changes.

Now we would say ___ because ___.

Switch speakers. In two dimensions, what must the force sums be along both axes for the forces to balance?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Stop the faulty graphic

19

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use all forces to calculate acceleration.

CHOOSE ONE JOB TOGETHER

SOFTWARE TESTER ☐

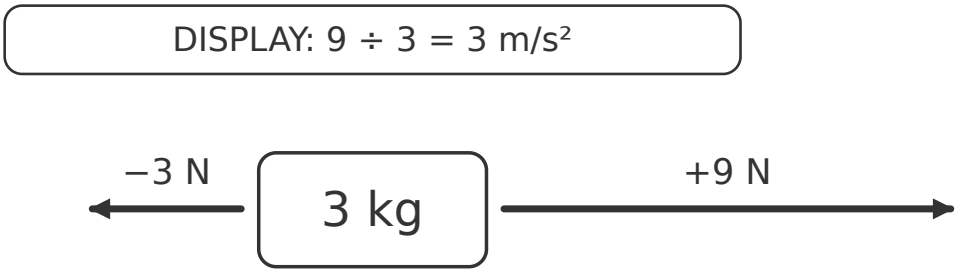
Mark the first wrong calculator step. Send the developer the corrected force sum and expected acceleration.

FACT-CHECK PRODUCER ☐

Stop or clear the graphic. Give the anchor a corrected number that viewers can check against the force arrows.

Word help: force sum / mass / acceleration

A calculator overlay for a **3 kg** tracker shows **+9 N** and **−3 N**. Its output is "**9 ÷ 3 = 3 m/s²**." It is about to go live.



01 OUR FIRST CALL

Can this calculation go live? Give the corrected result and explain.

The force sum is ___, so acceleration is ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Did the developer fix the error? Send a follow-up that names what still needs changing.

Now we would say ___ because ___.

Switch speakers. Create a different pair of opposing forces that would expose the same calculator error. Show the correct result.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Can we release a forecast?

20

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Identify which measurements a position forecast needs.

CHOOSE ONE JOB TOGETHER

MISSION SYSTEMS ENGINEER ☐

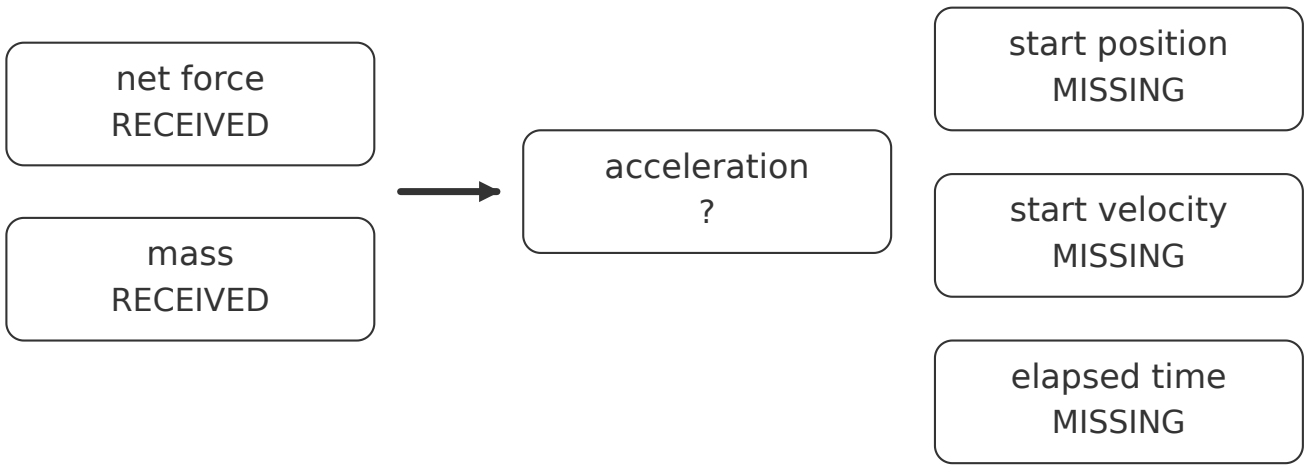
Mark what you can calculate now. Send the telescope operator a request for the measurements still needed.

NEWS ASSIGNMENT EDITOR ☐

Prepare a 'known / still needed' graphic. Tell the anchor which result can air and which must wait.

Word help: known / missing / forecast

A telescope manager wants a final position. The telemetry feed contains net force and mass, but no starting velocity, starting position or elapsed time.



01 OUR FIRST CALL

What can the desk release now? What measurements must the crew still collect?

We can find ___. We still need ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Is the forecast ready after the update? Say what you can release and what is still missing.

Now we would say ___ because ___.

Switch speakers. If XJ reverses direction during the forecast, how would that affect a report of distance versus displacement?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Two reports. One sighting.

21

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare position reports that use different zeros.

CHOOSE ONE JOB TOGETHER

SURVEY ENGINEER ☐

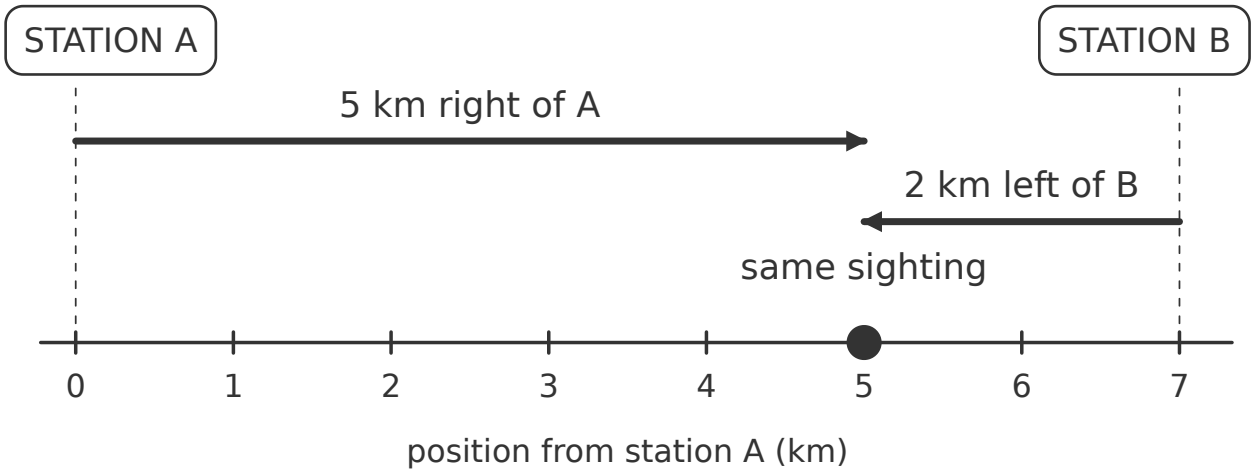
Use one axis for both stations. Give dispatch the position from A, then name what a future forecast still needs.

MAP BROADCASTER ☐

Show both stations and zeros on the locator map. Explain whether the two reports describe different locations.

Word help: zero / frame / position

Two observatories file the same sighting: "**5 km** right of station A" and "**2 km** left of station B." B is **7 km** right of A.



01 OUR FIRST CALL

Do the reports agree? Give one location and explain how the map supports it.

From station ___, the position is ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What number changes in the new frame? What physical location stays the same?

Now we would say ___ because ___.

Switch speakers. What measurement uncertainty should accompany a future central position forecast? Give a concrete example.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks

Would you air this headline?

22

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare forecast ranges and keep conclusions within the data.

CHOOSE ONE JOB TOGETHER

FORECAST ENGINEER ☐

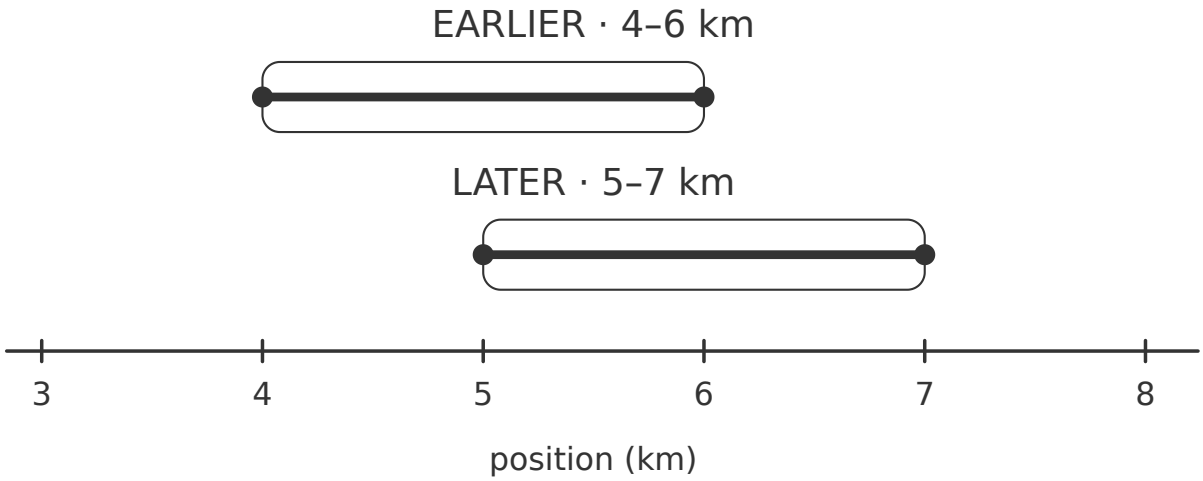
Mark both forecast bands. Issue a location bulletin and name an observation that would trigger another update.

SCIENCE NEWS EDITOR ☐

Decide whether to air the headline. Show both intervals and write the replacement line viewers should hear.

Word help: range / overlap / probability

Two short-interval forecasts in the same frame give **4-6 km** and **5-7 km**. The new estimate uses a later sighting. A headline says “Impact confirmed.”



01 OUR FIRST CALL

What location can you report? Should the proposed headline air? Explain.

The ranges show ___. They do not show ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Answer the caller using the bands. What can your desk report, and what is still unknown?

Now we would say ___ because ___.

Switch speakers. How is predicting XJ's location different from deciding what action people should take?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks